On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

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Q-1)

$$A=\begin{bmatrix}1 & \alpha & -\gamma & \alpha\\ 3\alpha & 5 & -9\gamma & 1\\ 2 & \beta & 3 & \alpha^2\\ 1 & 14 & -6 & 2\end{bmatrix}\begin{matrix}-8R_1+R_2\to R_2\\ -2R_1+R_3\to R_3\\ -R_1+R_4\to R_4\end{matrix}\begin{bmatrix}1 & \alpha & -\gamma & \alpha\\ 3\alpha-8 & 5-8\alpha & -9\gamma+8\gamma & 1-8\alpha\\ 0 & \beta-2\alpha & 3+2\gamma & \alpha^2-2\alpha\\ 0 & 14-2\alpha & -6+2\gamma & 2-\alpha\end{bmatrix}=\begin{bmatrix}1 & 8/3 & -\gamma & 8/3\\ 0 & -49/3 & -\gamma & -63/3\\ 0 & \beta-16/3 & 3+2\gamma & 16/9\\ 0 & 26/3 & -6+2\gamma & -2/3\end{bmatrix}$$

a)

$3\alpha-8=0$

$3\alpha=8$

$\alpha=8/3$ (m=2)

$\beta+11=0$

$\beta=-11$

$26-k.49=0$

$49k=26$

$k=\frac{26}{49}$

$-\frac{3\gamma}{98}-\frac{15}{2}=0$

$-\frac{3\gamma}{98}=\frac{15}{2}\Rightarrow\gamma=-245$

$$\begin{bmatrix}1 & 8/3 & -\gamma & 8/3\\ 0 & -49/3 & -\gamma & -63/3\\ 0 & 0 & 3+3\gamma & \frac{205}{9}\\ 0 & 0 & \frac{72}{49}\gamma-6 & \frac{248}{21}\end{bmatrix}=\begin{bmatrix}1 & 8/3 & -\gamma & 8/3\\ 0 & -49/3 & -\gamma & -63/3\\ 0 & \beta+11 & 3+3\gamma & 205/9\\ 0 & \frac{26}{3}-\frac{k.49}{3} & -6+2\gamma-k\gamma & \frac{-2}{3}-\frac{k.63}{3}\end{bmatrix}$$

$-R_2+R_3\to R_3$

$k.R_2+R_4\to R_4$

$$-\frac{R_3}{2}+R_4\to R_4$$

$$\begin{bmatrix}1 & 8/3 & \gamma & 8/3\\ 0 & -49/3 & -\gamma & -63/3\\ 0 & 0 & 3+3\gamma & \frac{205}{9}\\ 0 & 0 & -\frac{3\gamma}{98}-\frac{15}{2} & \frac{549}{126}\end{bmatrix}=\begin{bmatrix}1 & 8/3 & -245 & 8/3\\ 0 & -49/3 & 245 & -63/3\\ 0 & 0 & -732 & \frac{205}{9}\\ 0 & 0 & 0 & \frac{549}{126}\end{bmatrix}$$

b)

$$U=\begin{bmatrix}1 & 8/3 & -245 & 8/3\\ 0 & -49/3 & 245 & -21\\ 0 & 0 & -732 & 205/9\\ 0 & 0 & 0 & \frac{61}{14}\end{bmatrix}$$

c)

$$E_2=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & -1 & 1 & 0\\ 0 & \frac{26}{49} & 0 & 1\end{bmatrix},\quad E_1=\begin{bmatrix}1 & 0 & 0 & 0\\ -8 & 1 & 0 & 0\\ 2 & 0 & 1 & 0\\ -1 & 0 & 0 & 1\end{bmatrix},\quad E_3=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & -1/2 & 1\end{bmatrix}$$

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$\mathcal{E}_1$ ile 1. satırdaki pivotun altını sıfır yaptık;
$\mathcal{E}_2$ ile 2. satırdaki pivotun altını sıfır yaptık;
$\mathcal{E}_3$ ile 3. satırdaki pivotun altını sıfır yaptık ve A matrisini upper triangle matrix haline getirdik dolayısıyla başka dimensyon matrisi kalmamıştır.

Q-2)

Q-3) $H = I - 2u.u^T$

$$H_{nxn} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{nxn} - 2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{nxn} - 2 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{nxn}$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

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2) $H^2 = I$;

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}_{n\times n} \cdot \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}_{n\times n} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{n\times n}$$

b)

c) $|\lambda I - A| = 0$

$$H = \begin{vmatrix} \lambda+1 & 0 & 0 & 0 \\ 0 & \lambda+1 & 0 & 0 \\ 0 & 0 & \lambda+1 & 0 \\ 0 & 0 & 0 & \lambda+1 \end{vmatrix}$$

$$\det(H) = (\lambda+1)^n = 0$$

$$\lambda = -1$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad X = 0$$

d)

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Q-4) $V(0) = 10$ liters

$W(0) = 40$ liters

b) 'a'yı yapamadım $\frac{d}{dt}\begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} V_k \\ W_k \end{bmatrix}$ 'yı seçtim b ve c seçenekleri için.

$|\lambda I - A| = 0$

$$\begin{vmatrix} \lambda+2 & 1 \\ 1 & \lambda+2 \end{vmatrix} = 0$$

$(\lambda+2)^2 - 1 = 0$

$\lambda^2 + 4\lambda + 4 - 1 = 0$

$\lambda^2 + 4\lambda + 3 = 0$

$\lambda \quad +3$

$\lambda \quad +1$

$(\lambda+3)(\lambda+1)$

$\underline{\lambda = -3} \quad \underline{\lambda = -1}$

$\lambda = -3$ için;

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \cdot X_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-1 \cdot X_{11} + 1 \cdot X_{21} = 0$

$1 \cdot X_{11} - 1 \cdot X_{21} = 0$

$$X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = -1$ için;

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot X_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$1 \cdot X_{12} + 1 \cdot X_{21} = 0$

$1 \cdot X_{12} + 1 \cdot X_{21} = 0$

$$X_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} V_k \\ W_k \end{bmatrix} = C_1 \cdot \lambda_1 \cdot X_1 + C_2 \cdot \lambda_2 \cdot X_2 =$$

$$= C_1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-3)^n + C_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot (-1)^n$$

$n = 0$ için:

$$\begin{bmatrix} V_0 \\ W_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix} = C_1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$C_1 + C_2 = 1$

$C_1 - C_2 = 4$

$2C_1 = 5$

$C_1 = 5/2 \quad C_2 = -3/2$

$$\begin{bmatrix} V_k \\ W_k \end{bmatrix} = \frac{5}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-3)^k + \frac{-3}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot (-1)^k$$

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t=1 sec;

$$\begin{bmatrix} V_1 \\ W_1 \end{bmatrix} = \frac{5}{2}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot (-3)^1 - \frac{3}{2}\begin{bmatrix} 1 \\ -1 \end{bmatrix}(-1)^1 = \begin{bmatrix} -7.5 \\ -7.5 \end{bmatrix} + \begin{bmatrix} 1.5 \\ -1.5 \end{bmatrix}$$

%75 %15

c)

$h \to \infty$

$$\begin{bmatrix} V_\infty \\ W_\infty \end{bmatrix} = \begin{bmatrix} 7.5 \\ 1.5 \end{bmatrix}$$

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